Aim:

To prove carbon dioxide is released by germinating seeds during respiration.

Requirements:

A conical flask, cork, beaker, a twice bent glass tube, a small test tube, thread, water KOH, germinating seeds of bean / gram/ groundnut seeds.

Procedure :

1. Take a definite quantity (i.e 10 gm) of germinating seeds of bean/gram/groundnut in the conical flask and hang a small test tube containing Potassium hydroxide (KOH) crystal inside the flask with the help of a thread.
2. Introduce one end of the bent glass tube into the conical flask through the cork. Dip the free end of the tube in a beaker containing water.
3. Make the apparatus air tight and fix the apparatus with the help of a stand.
4. Note the initial level of water in the bent glass tube and keep the apparatus undisturbed.

Observation :

After two hours the level of water rises in the glass tube.

Inference :

Carbon dioxide released by the germinating seeds is absorbed by KOH solution. It creates vacuum, to fill up the vacuum water raised in the tube. Liberation of carbon dioxide during respiration by germinating seeds is proved.

Exercise: 36

Arc auxanometer experiment

Aim:

To measure the growth of a plant in length by Arc auxanometer

Requirements:

Arc auxanometer, potted plant, weight, thread,

Procedure:

Arc auxanometer which consists of a small pulley to the axis of which is attached a long pointer sliding over a graduated arc.

One end of a thread is tied to the stem tip and

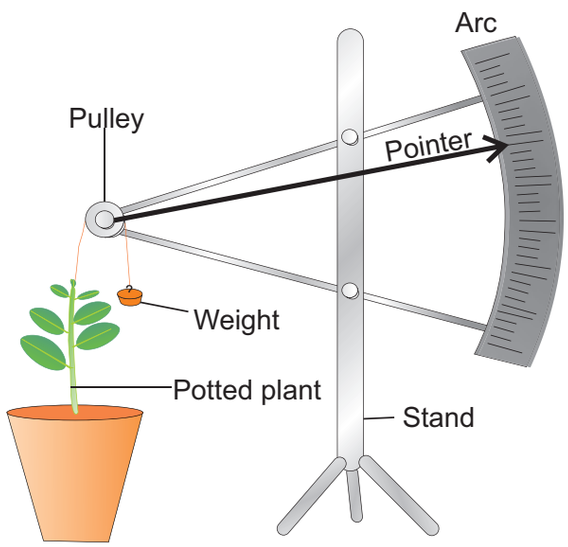


Figure 32 : Arc auxanometer experiment

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